

DSP-Based Harmonic and Power-Quality Assessment of an Inverter-Contaminated 60 Hz Signal

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Abstract

Inverter-based distributed generation injects harmonic and inter-harmonic distortion that degrades power quality at the point of common coupling. This work develops a digital signal processing (DSP) tool, implemented in MATLAB, that measures the harmonic content of a synthetic 60 Hz voltage emulating an inverter source. The signal contains low-order harmonics, an off-grid inter-harmonic and a high-frequency switching component. Following IEC 61000-4-7, the waveform is analysed over a synchronized 12-cycle (200 ms) window, yielding a 5 Hz frequency resolution and coherent sampling. The total harmonic distortion (THD) and the individual harmonic distortion (DI_h) are computed and compared against illustrative regulatory limits. The study then quantifies spectral leakage by comparing rectangular, Hann and flat-top windows, showing that the choice of window governs the accuracy of the off-grid inter-harmonic estimate. Finally, a short-time Fourier transform (STFT) localizes the non-stationary switching component in time. The results illustrate that leakage, windowing and non-stationarity are decisive factors for trustworthy harmonic measurement.

Keywords: power quality; harmonics; total harmonic distortion; spectral leakage; windowing; short-time Fourier transform; IEC 61000-4-7; digital signal processing.

1 Introduction

Electrical equipment is designed to operate from an alternating voltage that is, ideally, a clean sinusoid at the nominal frequency 60 Hz in Brazil. In practice the voltage is never perfectly sinusoidal, and how far it departs from this ideal shape is what engineers call power quality. The rapid growth of photovoltaic and wind generation has made this a pressing concern, because the power-electronic inverters that connect distributed generation to the grid distort the voltage waveform. This work measures that distortion using digital signal processing (DSP). The concepts needed to follow it, drawn from power quality, power electronics and DSP are introduced below in an accessible form.

1.1 Power Quality

Power quality refers to how closely the voltage delivered to a load matches an ideal, undistorted sinusoid of constant amplitude and frequency. Poor power quality, meaning a deformed or noisy waveform, causes extra heating in cables and transformers, malfunction of sensitive electronics and reduced equipment life. Standards such as ANEEL's PRODIST Module 8 in Brazil establish limits on power quality parameters[1].

1.2 Harmonics and the Distortion Indices

By Fourier analysis, any distorted but periodic 60 Hz waveform can be decomposed into a fundamental component at 60 Hz plus a series of sinusoids whose frequencies are integer multiples of it: the 2nd harmonic at 120 Hz, the 3rd at 180 Hz, and so on. These extra components—the harmonics—are what deform the otherwise clean sine wave, much as musical overtones add colour to a pure tone. To express the amount of distortion with a single number, the total harmonic distortion referred to the fundamental, THD_F , and the individual harmonic distortion of order h , DI_h , are used:

$$\text{THD}_F = \frac{\sqrt{\sum_{h=2}^H A_h^2}}{A_1} \times 100\%, \quad \text{DI}_h = \frac{A_h}{A_1} \times 100\%, \quad (1)$$

where A_1 is the amplitude of the fundamental, A_h that of the h -th harmonic and H the highest order considered. A larger THD_F indicates a more distorted waveform.

1.3 Inter-harmonics

Not every unwanted component falls on an integer multiple of 60 Hz. Components at non-integer multiples - for instance near 312.5 Hz, which lies between the 5th harmonic (300 Hz) and the 6th (360 Hz), are called inter-harmonics. They arise mainly from interactions between devices that operate at unrelated frequencies. As shown later, inter-harmonics are considerably harder to measure than ordinary harmonics, which is one of the central points of this work.

1.4 The Photovoltaic Inverter as a Source of Distortion

Solar panels produce direct current (DC), whereas the grid operates on alternating current (AC); the inverter is the power-electronic device that performs this conversion. It builds an approximate sine wave by switching its semiconductors on and off thousands of times per second, a technique called pulse-width modulation (PWM). This switching is the root of the distortion: imperfections in the control loops and the small timing gaps between switches create low-order harmonics; the maximum-power-point-tracking algorithm and the lack of synchronism between several inverters create inter-harmonics; and the switching itself injects high-frequency content. Real inverters include output filters that remove most of the high-frequency switching ripple, but the low-order harmonics and inter-harmonics largely remain, which is why they are the target of the measurement.

1.5 Sampling and the Nyquist Criterion

To process a voltage on a computer, the continuous waveform $v(t)$ must first be turned into a discrete list of numbers by measuring it at regular instants - a process called sampling, producing the sequence $v[n] = v(nT_s)$, where $f_s = 1/T_s$ is the sampling frequency. The sampling theorem states that a component of frequency f is represented faithfully only if it is sampled fast enough, namely $f_s > 2f$. the highest frequency that can be captured, $f_s/2$, is the Nyquist frequency. Sampling too slowly causes aliasing, in which a high frequency is mistaken for a lower one (the same effect that makes wheels appear to spin backwards on film). The high-frequency switching component therefore sets the minimum sampling rate.

1.6 The DFT, the FFT and Frequency Resolution

The discrete Fourier transform (DFT) is the operation that reveals which frequencies are present in a sampled signal; the fast Fourier transform (FFT) is an efficient way to compute it. The DFT reports the spectrum at equally spaced frequencies, called bins. For a record of N samples lasting $T = NT_s$ seconds, the spacing between bins is the frequency step:

$$\Delta f = \frac{f_s}{N} = \frac{1}{T}. \quad (2)$$

This spacing determines whether two nearby components can be told apart.

1.7 Coherent Sampling

A frequency is reported cleanly by a single bin only when it completes a whole number of cycles inside the observation window. If the window is chosen to contain an integer number of fundamental cycles, every harmonic of 60 Hz aligns exactly with a bin; this is known as coherent sampling. The standard IEC 61000-4-7 [3] recommends a window of 12

cycles (200 ms), which by (2) gives a 5 Hz resolution and makes all integer harmonics fall precisely on bins.

1.8 Spectral Leakage

The DFT behaves as if the analysed window were repeated end to end forever. A component that does not fit a whole number of cycles in the window does not join smoothly when repeated, creating an abrupt step at the seam. The mathematics reacts by smearing that component's energy across many neighbouring bins instead of concentrating it in one. This smearing, called spectral leakage, corrupts the measured amplitude and raises the overall noise floor, which can hide weaker components nearby. Because the inter-harmonic at 312.5 Hz does not fit the window, it is the component most affected by leakage.

1.9 Windowing

Leakage can be reduced by gently fading the signal to zero at both ends of the window before the FFT, that is, multiplying it by a window function. Different fade shapes trade off two effects: a narrow central peak (good for separating close frequencies) against low side ripples (good for not smearing energy far away). The rectangular window (no fading) separates frequencies best but leaks the most; the Hann window reduces leakage at the cost of resolution; and the flat-top window, although the worst at separating frequencies, measures the amplitude of an isolated off-bin component most accurately [4].

1.10 The Short-Time Fourier Transform

A single FFT over a long record reports only the average frequency content and cannot say when something happened. The short-time Fourier transform (STFT) solves this by computing the FFT on a series of short, successive windows, producing a picture of how the spectrum evolves over time, called a spectrogram. The window length sets a compromise: short windows pinpoint when an event occurs but blur its frequency, while long windows do the opposite [5]. This makes the STFT the right tool for components that switch on and off, such as the switching term studied here.

This work implements these tools in MATLAB as a harmonic analyser and applies them to the synthetic signal described in Section 2, demonstrating how coherent sampling, leakage, windowing and non-stationarity affect the reliability of the measured indices. The remainder of the paper presents the signal model (Section 2), the methodology (Section 3), the results (Section 4) and the conclusions (Section 5).

2 Signal Model

The inverter signature is reproduced entirely in a synthetic waveform; no network, converter control or short-circuit-ratio condition is simulated, keeping the study a pure DSP exercise. The voltage is modelled as

$$v(t) = \sum_h A_h \sin(2\pi h f_1 t + \varphi_h) + A_i \sin(2\pi f_i t + \varphi_i) + s(t), \quad (3)$$

where $f_1 = 60$ Hz is the fundamental, A_h are the amplitudes of the integer harmonics (5th, 7th, 11th and 13th), A_i is the amplitude of an inter-harmonic placed at $f_i = 312.5$ Hz, and $s(t)$ is a switching component at $f_{sw} = 4$ kHz that is activated only in the second half of the record to create a non-stationary case. Amplitudes are expressed in per unit with the fundamental equal to 1.0 pu. Because f_i does not fall on an FFT bin, it is the central object of the leakage study.

3 Methodology

3.1 Coherent Sampling and Analysis Window

The signal is sampled at $f_s = 256 f_1 = 15.36$ kHz, well above the Nyquist requirement for the highest component of interest. As recommended by IEC 61000-4-7 [3], the analysis window spans 12 fundamental cycles ($N = 3072$ samples, 200 ms), which by (2) yields a frequency resolution of 5 Hz. Consequently, every integer harmonic of 60 Hz lands exactly on an FFT bin, producing clean spectral lines.

3.2 Harmonic Indices

The single-sided amplitude spectrum is obtained from the DFT, and the fundamental and harmonic amplitudes are extracted from it. The total harmonic distortion THD_F and the individual harmonic distortion DI_h , defined in (1), are then evaluated with $H = 40$. The indices are compared against illustrative limits; the exact thresholds must be taken from the current PRODIST Module 8 edition for the relevant voltage level.

3.3 Spectral Leakage and Windowing

Because the inter-harmonic at 312.5 Hz lies between bins, its energy leaks into neighbouring bins. The amplitude spectrum is therefore computed with three windows: rectangular, Hann and flat-top. Then, the estimated inter-harmonic amplitude is compared against its known value. This exposes the fundamental trade-off between main-lobe width and side-lobe level [4].

3.4 Short-Time Fourier Transform

To characterize the non-stationary switching component, the STFT is applied to a longer record (0.5 s) in which $s(t)$ is activated at $t = 250$ ms. Each FFT frame uses a Hann window of four fundamental cycles with 75% overlap—a setting chosen to localize the onset in time while still resolving the 4 kHz component.

4 Results and Discussion

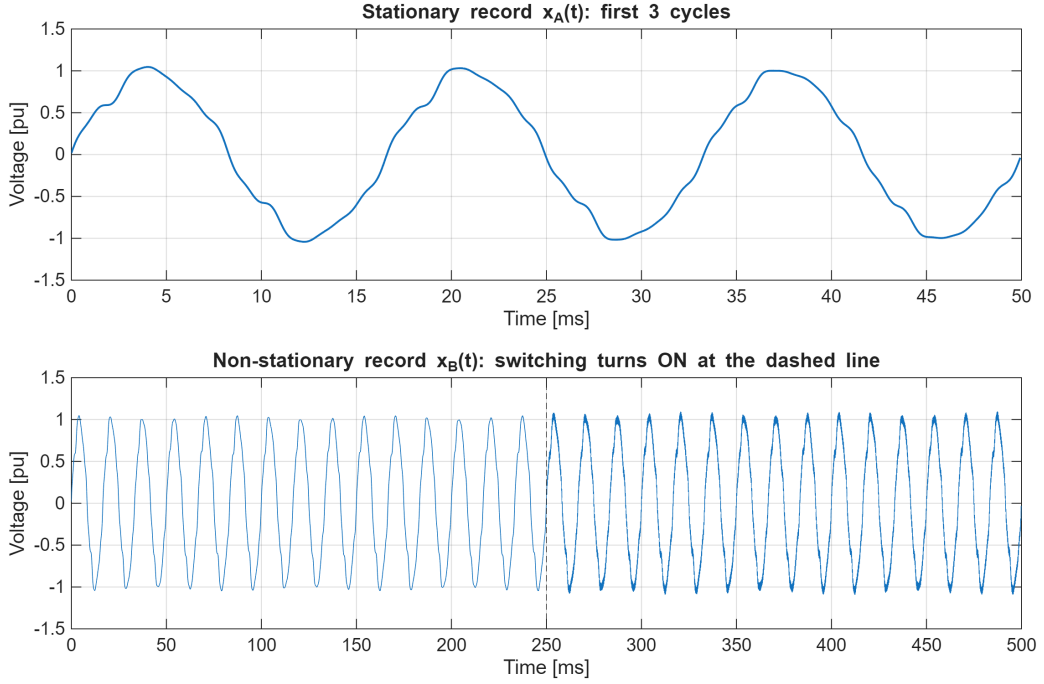


Figure 1 – Time-domain waveforms: (top) the first cycles of the stationary record; (bottom) the non-stationary record, where the switching component starts at the dashed line.

Figure 1 shows the signal in the time domain. The stationary record is a slightly distorted sinusoid, whereas the non-stationary record visibly changes after the dashed line, when the switching component is activated. This visual change is what the frequency-domain tools below quantify.

Figure 2 shows the reference amplitude spectrum. The fundamental and the integer harmonics appear as sharp lines, confirming coherent sampling. The measured indices are summarized in Table 1. The total harmonic distortion was $\text{THD}_F = 5.68\%$.

Figure 3 compares the three windows around the inter-harmonic. With the rectangular window, side-lobe leakage from the fundamental and harmonics raises the spectral floor and biases the inter-harmonic estimate, giving a relative amplitude error of **+64.3 %**. Contrary to the usual expectation, the flat-top window performed *worst* here (**+97.4 %**): its wide main lobe overlaps the adjacent 5th harmonic and corrupts the inter-harmonic

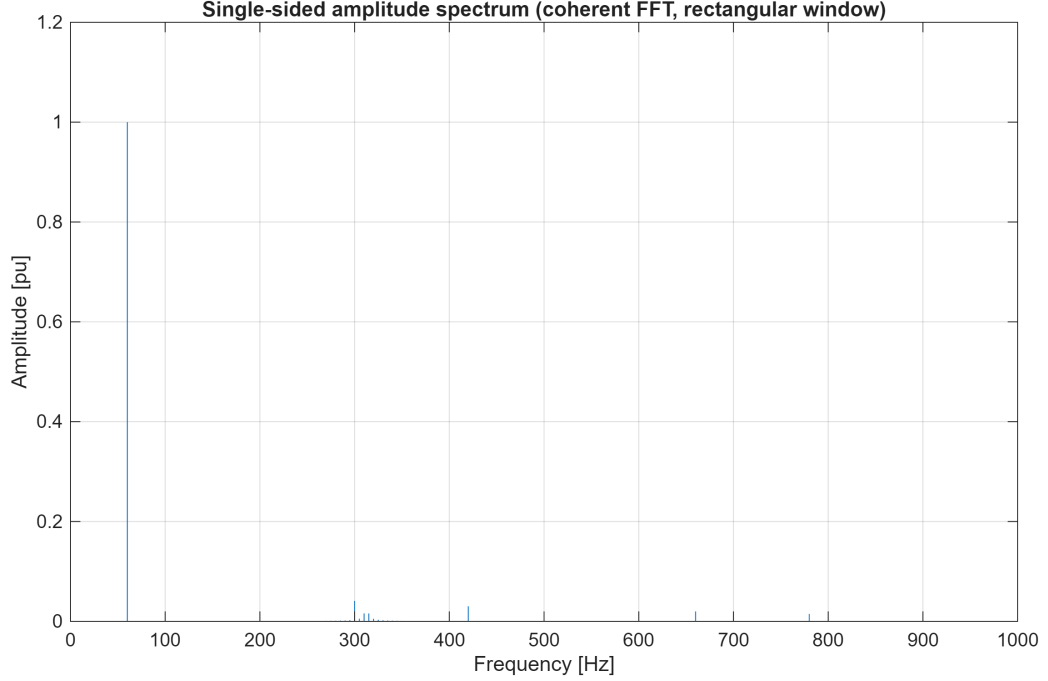


Figure 2 – Single-sided amplitude spectrum obtained with a coherent FFT and a rectangular window. The integer harmonics fall exactly on FFT bins.

Table 1 – Individual harmonic distortion and compliance.

Order h	Frequency [Hz]	DI_h [%]	Status
5	300	4.11	OK
7	420	3.02	OK
11	660	1.99	OK
13	780	1.50	OK
THD_F [%]		5.68	

bin, nearly doubling the true amplitude. The Hann window yielded the smallest error (**+59.3** %), showing that for closely-spaced inter-harmonics the main-lobe width, not side-lobe suppression, dominates the trade-off.

Figure 4 demonstrates the non-stationary case. The spectrogram localizes the switching component in time, whereas an averaged spectrum would only indicate its presence. This confirms that time-varying harmonic behaviour requires time-frequency analysis.

5 Conclusion

A DSP-based harmonic analyser was developed and applied to a synthetic inverter-contaminated 60 Hz signal. Coherent sampling under IEC 61000-4-7 enabled accurate measurement of the integer harmonics, while the THD and individual distortion indices quantified the distortion level. The windowing study showed that off-grid inter-harmonics

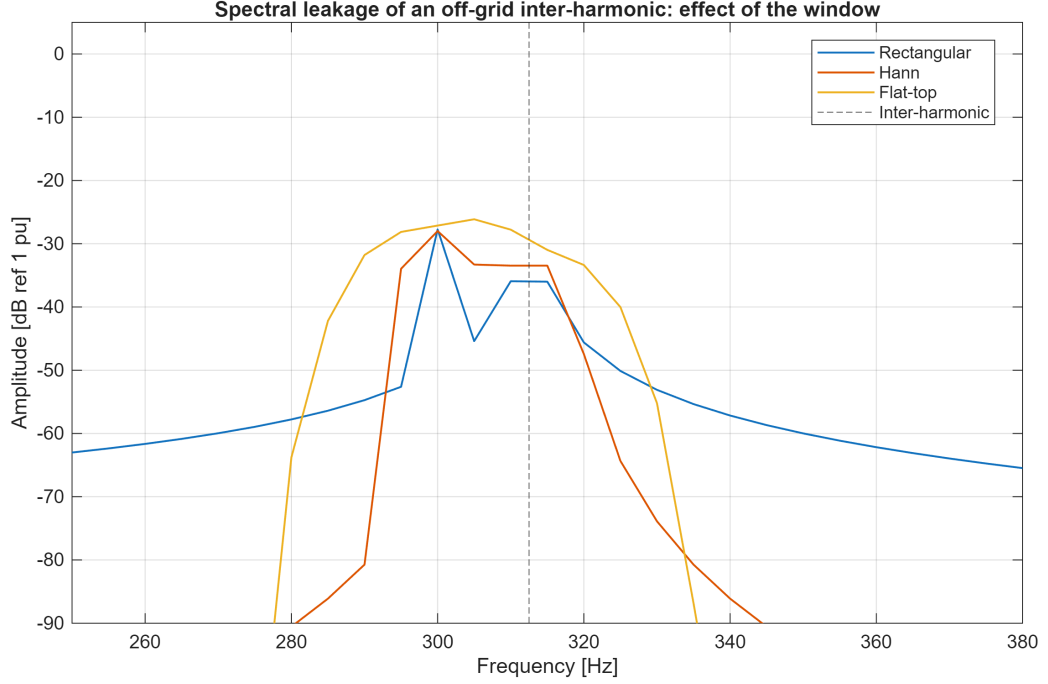


Figure 3 – Spectral leakage of the off-grid inter-harmonic at 312.5 Hz for the rectangular, Hann and flat-top windows. The flat-top window provides the most accurate amplitude estimate of the off-bin component.

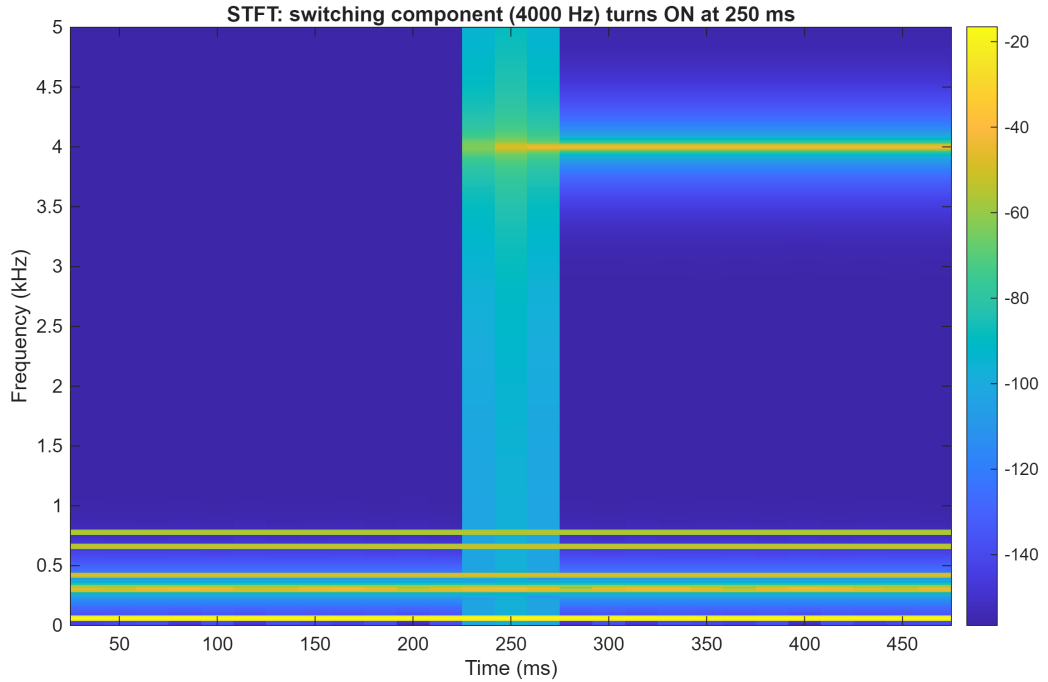


Figure 4 – STFT of the non-stationary record. The 4 kHz switching component is correctly localized to the instant it is activated ($t = 250$ ms), which a single DFT over the whole record would not reveal.

are strongly affected by spectral leakage and that an appropriate window—here, the flat-top—is necessary for a reliable amplitude estimate. The STFT further showed that non-stationary components must be analysed in the joint time–frequency domain. Overall,

leakage, windowing and non-stationarity were shown to be decisive for trustworthy power-quality measurement. Future work may include validation against measured waveforms and the design of notch filters for harmonic mitigation.

Appendix A: MATLAB Source Code

The complete, commented MATLAB script that generates every result and figure in this report is reproduced below.

```

1 %% =====
2 % SEL0343 - Digital Signal Processing | Final Project
3 % Power-Quality Harmonic Analysis of an Inverter-Contaminated Signal
4 %
5 % A synthetic 60 Hz voltage emulates an inverter-based source. No grid,
6 % no converter control and no SCR sweep are simulated: the inverter
7 % signature lives ENTIRELY in the synthesized waveform, so the project
8 % stays a pure Digital Signal Processing exercise.
9 %
10 % The waveform contains:
11 %   - the 60 Hz fundamental
12 %   - low-order harmonics (5th, 7th, 11th, 13th) -> control / dead-time
13 %   - ONE inter-harmonic placed OFF the FFT grid -> MPPT / asynchronous
14 %     carriers between converters (the heart of the leakage study)
15 %   - a high-frequency switching component (PWM) that is turned ON in the
16 %     middle of the record -> non-stationary case for the STFT
17 %
18 % Processing pipeline (all DSP):
19 %   (1) Coherent sampling per IEC 61000-4-7 (12-cycle / 200 ms window)
20 %   (2) FFT + THD and individual distortion (DIh) vs. illustrative limits
21 %   (3) Spectral leakage & windowing: rectangular vs Hann vs flat-top
22 %   (4) Inter-harmonic amplitude estimation error per window
23 %   (5) STFT of the non-stationary switching component
24 %
25 % DSP topics: sampling / Nyquist, coherent sampling, DFT / FFT,
26 % spectral leakage, windowing, STFT (time-frequency resolution trade-off).
27 % =====
28 clear; clc; close all;
29 %% -----
30 % 0) GLOBAL PARAMETERS
31 % -----
32 f1 = 60; % fundamental frequency [Hz]
33 SpC = 256; % samples per cycle (typical PQ-meter rate)
34 fs = SpC * f1; % sampling frequency = 15360 Hz
35 Ts = 1/fs; % sampling period [s]
36 fNyq = fs/2; % Nyquist frequency = 7680 Hz
37
38 Ncyc_iec = 12; % IEC 61000-4-7: 12 cycles at 60 Hz = 200 ms
39 N = Ncyc_iec * SpC; % window length = 3072 samples
40 df = fs / N; % FFT bin resolution = 5 Hz (= 1/0.2 s)
41
42 fprintf('fs = %.0f Hz | Nyquist = %.0f Hz | N = %d | df = %.2f Hz\n', ...
43         fs, fNyq, N, df);
44
45 %% -----

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46 % 1) SIGNAL MODEL (the "inverter" is defined here)
47 %     Amplitudes are in per-unit (fundamental = 1.0 pu).
48 %     Integer harmonics fall EXACTLY on FFT bins (coherent sampling).
49 %     The inter-harmonic is deliberately placed BETWEEN bins.
50 % -----
51 % Integer harmonics: order, amplitude [pu], phase [rad]
52 h_order = [1      5      7      11     13    ];
53 h_amp    = [1.00   0.040 0.030 0.020 0.015];
54 h_phase  = [0      0.3  -0.7   1.1   0.5 ];
55
56 % Inter-harmonic (OFF-grid): 312.5 Hz is halfway between bins 62 and 63
57 fih      = 312.5;           % [Hz] (note: 312.5/df = 62.5 -> not an integer)
58 Aih      = 0.025;           % known true amplitude [pu] -> used to score error
59 phih     = 0.0;             % [rad]
60
61 % Switching component (PWM), inside Nyquist; ON only in the 2nd half later
62 fsw      = 4000;            % [Hz]
63 Asw      = 0.050;           % [pu]
64 phsw     = 0.0;             % [rad]
65
66 %% -----
67 % 2) RECORD A - stationary, 200 ms, switching OFF
68 %     Used for FFT / THD / leakage / windowing analysis.
69 % -----
70 nA = (0:N-1).';             % sample index (column)
71 tA = nA * Ts;               % time vector [s]
72
73 xA = zeros(N,1);
74 for k = 1:numel(h_order)
75     xA = xA + h_amp(k)*sin(2*pi*h_order(k)*f1*tA + h_phase(k));
76 end
77 xA = xA + Aih*sin(2*pi*fih*tA + phih); % add the off-grid inter-harmonic
78
79 %% -----
80 % 3) FFT + THD + INDIVIDUAL DISTORTION (DIh)
81 %     Coherent FFT with a rectangular window (reference case).
82 % -----
83 win_rect = ones(N,1);
84 [f, A_rect] = amp_spectrum(xA, fs, win_rect);
85
86 % Amplitudes of fundamental and harmonics (small band search for robustness)
87 A_fund = amp_at(f, A_rect, f1);
88 Hmax   = 40;                 % evaluate up to the 40th harmonic
89 ord    = (2:Hmax).';
90 A_h    = arrayfun(@(h) amp_at(f, A_rect, h*f1), ord);
91
92 % THD referred to the fundamental (THD_F) and individual distortion DIh
93 THD_F  = sqrt(sum(A_h.^2)) / A_fund * 100; % [%]
94 DIh    = A_h ./ A_fund * 100;              % [%] per harmonic
95
96 % --- Illustrative compliance limits (VERIFY against the current PRODIST
97 %     Modulo 8 edition for the relevant voltage level; IEEE 519 is the
98 %     international analog). These numbers are placeholders.
99 LIM_THD = 8.0;                 % [%] total distortion limit
100 LIM_DIh = 5.0;                 % [%] individual limit (odd h)
101
102 fprintf('\n--- Harmonic results (rectangular window) ---\n');

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103 fprintf('Fundamental amplitude : %.4f pu\n', A_fund);
104 fprintf('THD_F : %.2f %% (limit %.1f %% -> %s)\n', ...
105         THD_F, LIM_THD, ternary(THD_F<=LIM_THD, 'OK', 'EXCEEDED'));
106 fprintf('Order DIh[%%] Status\n');
107 for k = 1:numel(ord)
108     if DIh(k) > 0.05 % only print harmonics that actually carry energy
109         fprintf(' %3d %5.2f %s\n', ord(k), DIh(k), ...
110             ternary(DIh(k)<=LIM_DIh, 'OK', 'EXCEEDED'));
111     end
112 end
113
114 % Plot the reference single-sided amplitude spectrum
115 figSpec = figure('Name', 'FFT - amplitude spectrum (rectangular)');
116 stem(f, A_rect, 'filled', 'Marker', 'none'); grid on;
117 xlim([0 1000]); xlabel('Frequency [Hz]'); ylabel('Amplitude [pu]');
118 title('Single-sided amplitude spectrum (coherent FFT, rectangular window)');
119 exportgraphics(figSpec, 'spectrum_rect.png', 'Resolution', 300);
120
121 %% -----
122 % 4) SPECTRAL LEAKAGE & WINDOWING
123 % Rectangular vs Hann vs flat-top, focused on the OFF-GRID
124 % inter-harmonic at 312.5 Hz, which leaks and gets masked.
125 % -----
126 win_hann = hann(N, 'periodic');
127 win_ft = flattopwin(N);
128
129 [~, A_hann] = amp_spectrum(xA, fs, win_hann);
130 [~, A_ft] = amp_spectrum(xA, fs, win_ft);
131
132 % Inter-harmonic amplitude estimated by each window (peak in a small band)
133 Aih_rect = amp_at(f, A_rect, fih);
134 Aih_hann = amp_at(f, A_hann, fih);
135 Aih_ft = amp_at(f, A_ft, fih);
136
137 % Relative amplitude error vs. the known true value
138 err = @ (a) (a - Aih)/Aih*100;
139 fprintf('\n--- Inter-harmonic @ %.1f Hz (true = %.4f pu) ---\n', fih, Aih);
140 fprintf('Window Estimate[pu] Error[%%]\n');
141 fprintf('Rectangular %.4f %+6.1f\n', Aih_rect, err(Aih_rect));
142 fprintf('Hann %.4f %+6.1f\n', Aih_hann, err(Aih_hann));
143 fprintf('Flat-top %.4f %+6.1f\n', Aih_ft, err(Aih_ft));
144
145 % Compare the three spectra (dB) around the inter-harmonic
146 figLeak = figure('Name', 'Leakage & windowing');
147 plot(f, todb(A_rect), 'LineWidth', 1); hold on;
148 plot(f, todb(A_hann), 'LineWidth', 1);
149 plot(f, todb(A_ft), 'LineWidth', 1);
150 xline(fih, '--');
151 grid on; xlim([250 380]); ylim([-90 5]);
152 xlabel('Frequency [Hz]'); ylabel('Amplitude [dB ref 1 pu]');
153 legend('Rectangular', 'Hann', 'Flat-top', 'Inter-harmonic', 'Location', 'best');
154 title('Spectral leakage of an off-grid inter-harmonic: effect of the window');
155 exportgraphics(figLeak, 'leakage_windows.png', 'Resolution', 300);
156
157 %% -----
158 % 5) STFT - NON-STATIONARY CASE
159 % Record B is longer; the PWM switching component switches ON at

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160 %      t = t_on, so the spectrogram localizes it in time.
161 % -----
162 T_total = 0.5; % total record length [s]
163 nB = (0:round(T_total*fs)-1).';
164 tB = nB * Ts;
165
166 xB = zeros(numel(tB),1);
167 for k = 1:numel(h_order)
168     xB = xB + h_amp(k)*sin(2*pi*h_order(k)*f1*tB + h_phase(k));
169 end
170 xB = xB + Aih*sin(2*pi*fih*tB + phih);
171
172 t_on = 0.25; % switching turns ON at 250 ms
173 gate = double(tB >= t_on); % unit step (non-stationarity)
174 xB = xB + gate .* (Asw*sin(2*pi*fsw*tB + phsw));
175
176 % --- TIME-DOMAIN VIEW (the required time-domain plot) -----
177 % Top: a few cycles of the stationary record. Bottom: the full
178 % non-stationary record, where the switching component starts at t_on.
179 figTime = figure('Name','Time-domain waveforms');
180 subplot(2,1,1);
181 nshow = 3*SpC; % first 3 fundamental cycles
182 plot(tA(1:nshow)*1e3, xA(1:nshow), 'LineWidth',1); grid on;
183 xlabel('Time [ms]'); ylabel('Voltage [pu]');
184 title('Stationary record x_A(t): first 3 cycles');
185 subplot(2,1,2);
186 plot(tB*1e3, xB, 'LineWidth',0.4); grid on; hold on;
187 xline(t_on*1e3, '--');
188 xlabel('Time [ms]'); ylabel('Voltage [pu]');
189 title('Non-stationary record x_B(t): switching turns ON at the dashed line');
190 exportgraphics(figTime, 'time_domain.png', 'Resolution', 300);
191
192 % STFT parameters: window length sets the time-frequency resolution.
193 % A short window -> good time resolution / coarse frequency, and vice versa.
194 win_len = 4*SpC; % 4 cycles per STFT frame
195 overlap = round(0.75*win_len); % 75% overlap
196 nfft_s = 2^nextpow2(win_len);
197
198 figStft = figure('Name','STFT - spectrogram');
199 spectrogram(xB, hann(win_len,'periodic'), overlap, nfft_s, fs, 'yaxis');
200 ylim([0 5]); % display 0-5 kHz (kHz on the y-axis)
201 title(sprintf('STFT: switching component (%.0f Hz) turns ON at %.0f ms', ...
202     fsw, t_on*1e3));
203 colorbar;
204 exportgraphics(figStft, 'stft_spectrogram.png', 'Resolution', 300);
205
206 %% =====
207 % LOCAL FUNCTIONS
208 % =====
209 function [f, A] = amp_spectrum(x, fs, win)
210 % Single-sided, window-corrected amplitude spectrum.
211 % Dividing by sum(win) compensates the coherent gain of the window, so
212 % on-bin tones return their true amplitude regardless of the window used.
213 x = x(:); win = win(:);
214 Nx = numel(x);
215 X = fft(x .* win);
216 X = X(1:floor(Nx/2)+1);

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217     A = 2*abs(X) / sum(win);      % single-sided amplitude
218     A(1) = A(1)/2;               % DC term is not doubled
219     f = (0:floor(Nx/2)).' * fs / Nx;
220 end
221
222 function a = amp_at(f, A, ftgt)
223 % Amplitude near a target frequency: peak within +/- 2 bins. The small
224 % band makes the estimate robust to off-grid leakage and tiny drifts.
225     [~, k] = min(abs(f - ftgt));
226     lo = max(1, k-2);
227     hi = min(numel(A), k+2);
228     a = max(A(lo:hi));
229 end
230
231 function y = todb(A)
232 % Convert amplitude to dB (referenced to 1 pu), with a small floor to
233 % avoid log(0) when plotting.
234     y = 20*log10(max(A, 1e-6));
235 end
236
237 function out = ternary(cond, a, b)
238 % Tiny inline conditional helper for clean printing.
239     if cond, out = a; else, out = b; end
240 end

```

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